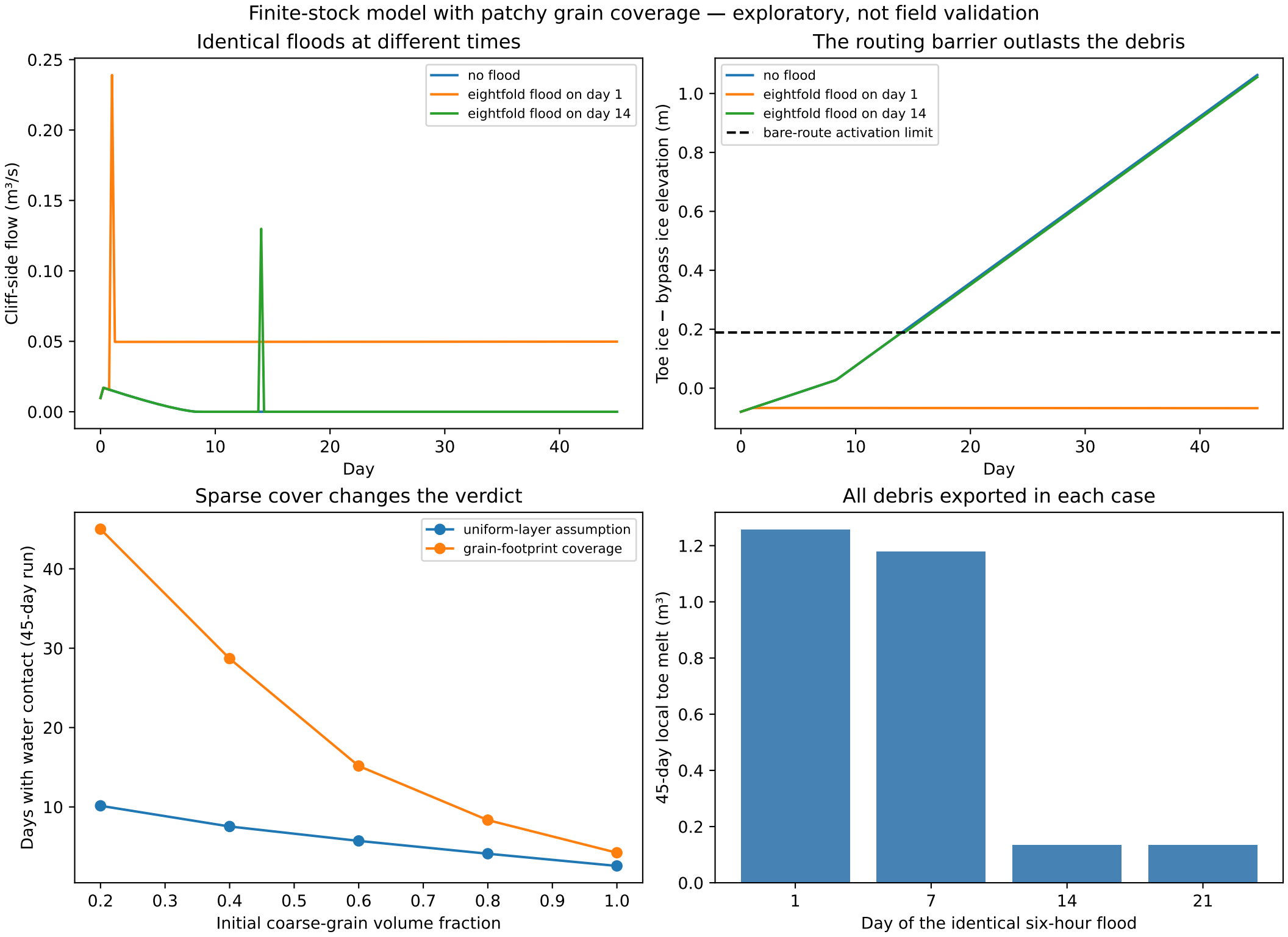
Finite-stock model with patchy grain coverage — exploratory, not field validation
Identical floods at different times
Cliff-side flow (m³/s)
Day
no flood
eightfold flood on day 1
eightfold flood on day 14
The routing barrier outlasts the debris
Toe ice – bypass ice elevation (m)
Day
no flood
eightfold flood on day 1
eightfold flood on day 14
bare-route activation limit
Sparse cover changes the verdict
Days with water contact (45-day run)
Initial coarse-grain volume fraction
uniform-layer assumption
grain-footprint coverage
All debris exported in each case
45-day local toe melt (m³)
Day of the identical six-hour flood